

# PLIP Interaction Report

**Compound:** (2R,3Z)-Phycocyanobilin | **Receptor:** 1MJE (mouse BRCA2 DBD, chain A) | **Best ΔG:** -9.425 kcal/mol  
Source: PLIP (plip-tool.biotec.tu-dresden.de) — Track 1, 1MJE receptor

## Hydrophobic Interactions - - - - -

| Index | Residue | AA  | Distance (Å) | Ligand Atom | Protein Atom |
|-------|---------|-----|--------------|-------------|--------------|
| 1     | 2453A   | PRO | 3.60         | 43          | 606          |
| 2     | 2550A   | TRP | 3.89         | 15          | 1515         |
| 3     | 2551A   | LYS | 3.75         | 10          | 1522         |
| 4     | 2554A   | ALA | 3.65         | 35          | 1550         |
| 5     | 2558A   | ALA | 3.90         | 44          | 1587         |
| 6     | 2559A   | PHE | 3.78         | 43          | 1598         |
| 7     | 2670A   | GLN | 3.55         | 25          | 2468         |
| 8     | 2709A   | TRP | 3.72         | 43          | 2824         |

## Hydrogen Bonds —

| Index | Residue | AA  | Dist H-A (Å) | Dist D-A (Å) | Donor Angle (°) | Protein donor? | Side chain | Donor Atom | Acceptor Atom |
|-------|---------|-----|--------------|--------------|-----------------|----------------|------------|------------|---------------|
| 1     | 2671A   | LYS | 1.97         | 2.91         | 151.95          | No (ligand)    | ×          | 2474 [Nam] | 41 [O.co2]    |

Note: Strongest geometric hydrogen bond of all four compounds (D-A 2.91 Å, angle 151.95°). Ligand is the donor.